HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2026)

Student No.: 22 Name: _____ Marks: / 67

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. In Figure 1, ABC is a triangle. D is a point lying on AB. Let E and F be points lying on CB produced and CD produced respectively such that $BD//EF$ and $BC = EF$. It is given that $BD = 8$ cm and $BE = 6$ cm.

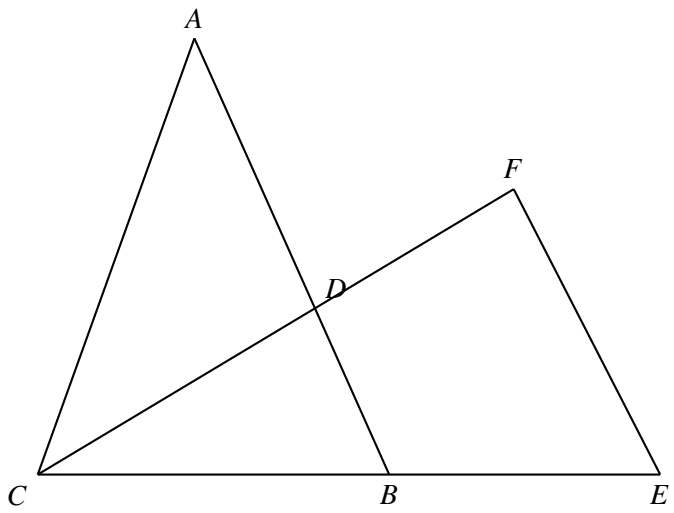


Figure 1

(a) By considering a pair of similar triangles, find the length of BC.

(b) Suppose that $AD = 10$ cm. Prove that $\triangle ABC \cong \triangle CEF$.

(5 marks) (DSE-2026 PI Q08)

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2. The stem-and-leaf diagram below shows the distribution of the numbers of driving hours of a group of taxi drivers in a week.

Stem (tens)	Leaf (units)
2	5 6 9
3	2 3 3 6 6 7 8
4	0 0 2 2 5 5 7 7 7
5	w

The ratio of the range to the inter-quartile range of the distribution is 7 : 3.

(a) Find w.

(b) (i) Write down the mode of the distribution.

(ii) If two taxi drivers are randomly selected from the group, find the probability that the mean of their numbers of driving hours in the week exceeds the mode of the distribution.

(6 marks) (DSE-2026 PI Q10)

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3. The straight line L passes through the point $(-10, 11)$. The slope of L is -2.

(a) Find the equation of L.

(b) The coordinates of the points H and K are $(2, 5)$ and $(-4, -7)$ respectively. Let P be a point lying on L such that P is equidistant from H and K.

(i) Find the coordinates of P.

(ii) Let Q be a point such that $HPKQ$ is a quadrilateral, where $HP = HQ = KQ$. Find the area of the quadrilateral $HPKQ$.

(7 marks) (DSE-2026 PI Q12)

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4. A bag contains 3 red balls, 3 white balls and 5 black balls. If 4 balls are randomly drawn from the bag at the same time, find

(a) the probability that exactly 2 red balls are drawn;

(b) the probability that exactly 2 red balls or exactly 2 white balls are drawn.

(4 marks) (DSE-2026 PI Q15)

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5. In Figure 2, the base of the pyramid $VABCD$ is the quadrilateral $ABCD$. The face VAB of the pyramid lies on the horizontal ground. It is given that $AB = 90$ cm, $AD = 70$ cm, $CD = 100$ cm, $\angle ADC = 55°$ and $\angle BCD = 130°$.

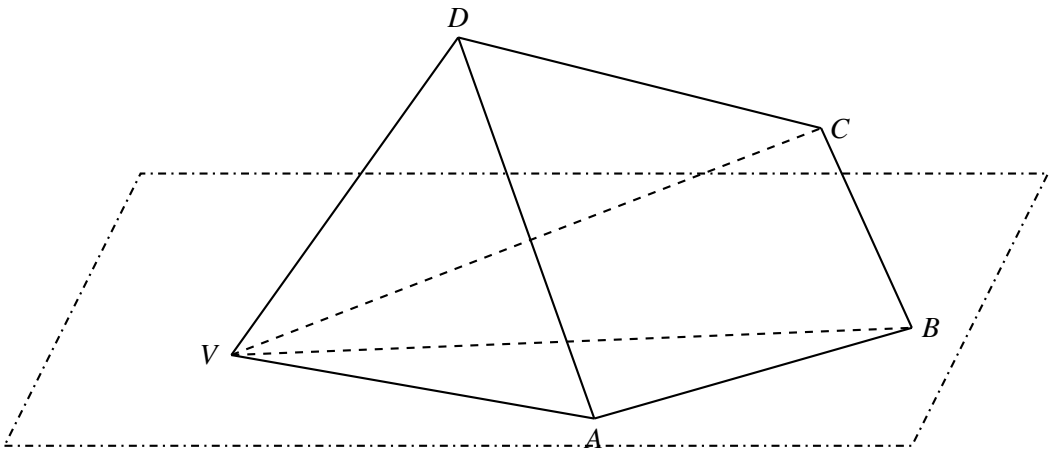


Figure 2

(a) Find

(i) the distance between A and C,

(ii) $\angle BAD$.

(b) It is given that $VD = 80$ cm. A student finds that the angle between the plane $ABCD$ and the horizontal ground is 65°. The student claims that the angle between VD and the horizontal ground is less than 50°. Is the claim correct? Explain your answer.

(6 marks) (DSE-2026 PI Q17)

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6. Let $G(n)$ be the nth term of a geometric sequence. It is given that $G(2) = 12$ and $G(5) = 768$.

(a) Find $G(1)$.

(b) Define $T(n) = \log_2 G(n)$ for all positive integers n. Find

(i) $T(n+1) - T(n)$,

(ii) the greatest value of k such that $T(1) + T(2) + T(3) + \cdots + T(k) < 10^6$.

(7 marks) (DSE-2026 PI Q18)

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7. Let $h(x) = -x^2 + 30kx + 4x - 225k^2 - 40k + 3$, where k is a positive constant. On the same rectangular coordinate system, denote the vertex of the graph of $y = h(x)$ and the vertex of the graph of $y = 14 - h(4-x)$ by A and B respectively.

(a) Using the method of completing the square, express, in terms of k, the coordinates of A.

(b) Write down, in terms of k, the coordinates of B.

(c) The coordinates of the point C are $(2 + 48k,\ 7 - 36k)$. Let I and J be the in-centre and the circumcentre of $\triangle ABC$ respectively.

(i) Let M be the mid-point of AB. Describe the geometric relationship between CM and AB. Explain your answer.

(ii) Express, in terms of k, the distance between I and J.

(iii) The coordinates of the point D are $(6 - 22k,\ 6 - 15k)$. Is it possible that $BDIJ$ is a parallelogram? Explain your answer.

(12 marks) (DSE-2026 PI Q19)

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8. Let a_n be the nth term of a sequence. If $a_2 = 7$, $a_5 = 35$ and $a_{n+2} = 3a_{n+1} - 2a_n$ for any positive integer n, then $a_8 =$

A. 67.

B. 131.

C. 259.

D. 515.

(DSE-2026 PII Q13)

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9. The perimeter and the area of a sector are k cm and $3k$ cm^2 respectively. If the radius of the sector is 14 cm, then the arc length of the sector is

A. 19 cm.

B. 21 cm.

C. 28 cm.

D. 49 cm.

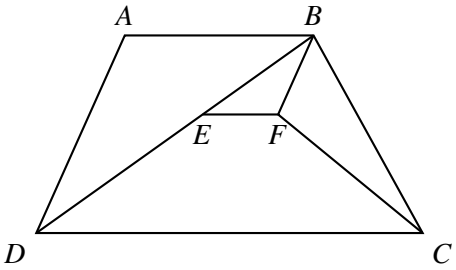
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10. In the figure, $ABCD$ is a trapezium with $AB//DC$ and $AB : CD = 3 : 5$. Let E be a point lying on BD such that $BE : ED = 2 : 3$. It is given that $AB//EF$ and $AD//BF$. If the area of $\triangle ABD$ is 225 cm^2, then the area of the trapezium $CDEF$ is



A. 222 cm^2.

B. 260 cm^2.

C. 279 cm^2.

D. 285 cm^2.

(DSE-2026 PII Q18)

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11. The point $(3, 1)$ is translated leftwards by 5 units to the point U. U is then rotated anticlockwise about the origin through $270°$ to the point V. Find the y-coordinate of V.

A. -2

B. -1

C. 1

D. 2

(DSE-2026 PII Q24)

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12. It is given that $\log y$ is a linear function of x. The intercepts on the vertical axis and on the horizontal axis of the graph of the linear function are 2 and 1 respectively. If $y = ab^x$, find a.

A. 0.01

B. 0.1

C. 10

D. 100

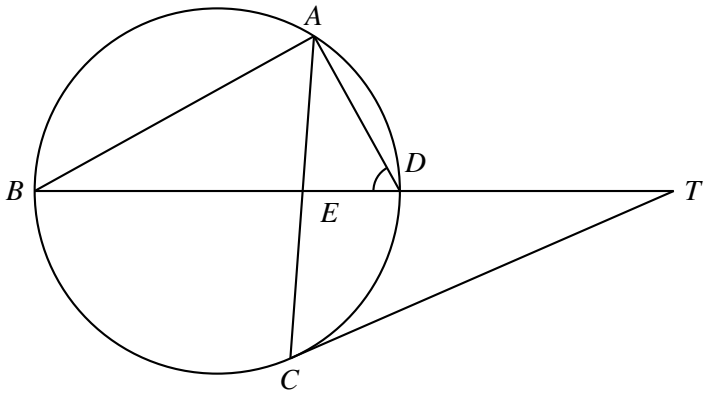
(DSE-2026 PII Q36)

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13. In the figure, $ABCD$ is a circle. AC and BD intersect at the point E. BD is produced to the point T such that TC is tangent to the circle. It is given that $AB//TC$. If $\angle AED = 85°$ and $\angle CTE = 23°$, then $\angle ADE =$



A. 39°.

B. 56°.

C. 62°.

D. 72°.

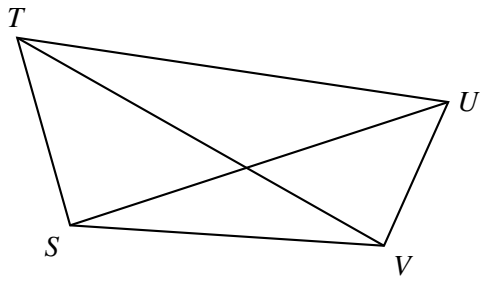
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14. The figure shows the quadrilateral $STUV$, where $ST = 8$ cm and $SV = 5$ cm. If $\angle UTV = 30^\circ$ and $\angle TSU = \angle TVU = 90^\circ$, find $\angle SUV$ correct to the nearest degree.



A. 22°

B. 28°

C. 30°

D. 38°

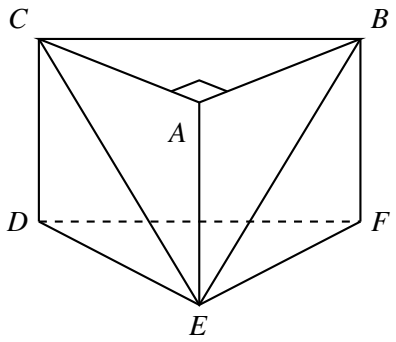
(DSE-2026 PII Q39)

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15. The figure shows the right prism $ABCDEF$, where $AB : AC : AE = 1 : 1 : 2$. Let θ be the angle between the plane ABC and the plane BCE. If $\angle BAC = 90°$, find $\sin\theta$.



A. $\frac{1}{3}$

B. $\frac{2}{3}$

C. $\frac{\sqrt{2}}{4}$

D. $\frac{2\sqrt{2}}{3}$

(DSE-2026 PII Q40)

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16. In a class, there are 3 boys and 7 girls. If 5 students are randomly selected from the class to form a committee, find the probability that at most 2 boys are selected.

A. $\frac{5}{6}$

B. $\frac{11}{12}$

C. $\frac{119}{120}$

D. $\frac{251}{252}$

(DSE-2026 PII Q42)

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17. There are 5 full-time employees and 3 part-time employees in a company. From the company, 7 employees are selected to form a queue. If no part-time employees are next to each other, how many different queues can be formed?

A. 5040

B. 14400

C. 18000

D. 32400

(DSE-2026 PII Q43)

I can finish _____ questions

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